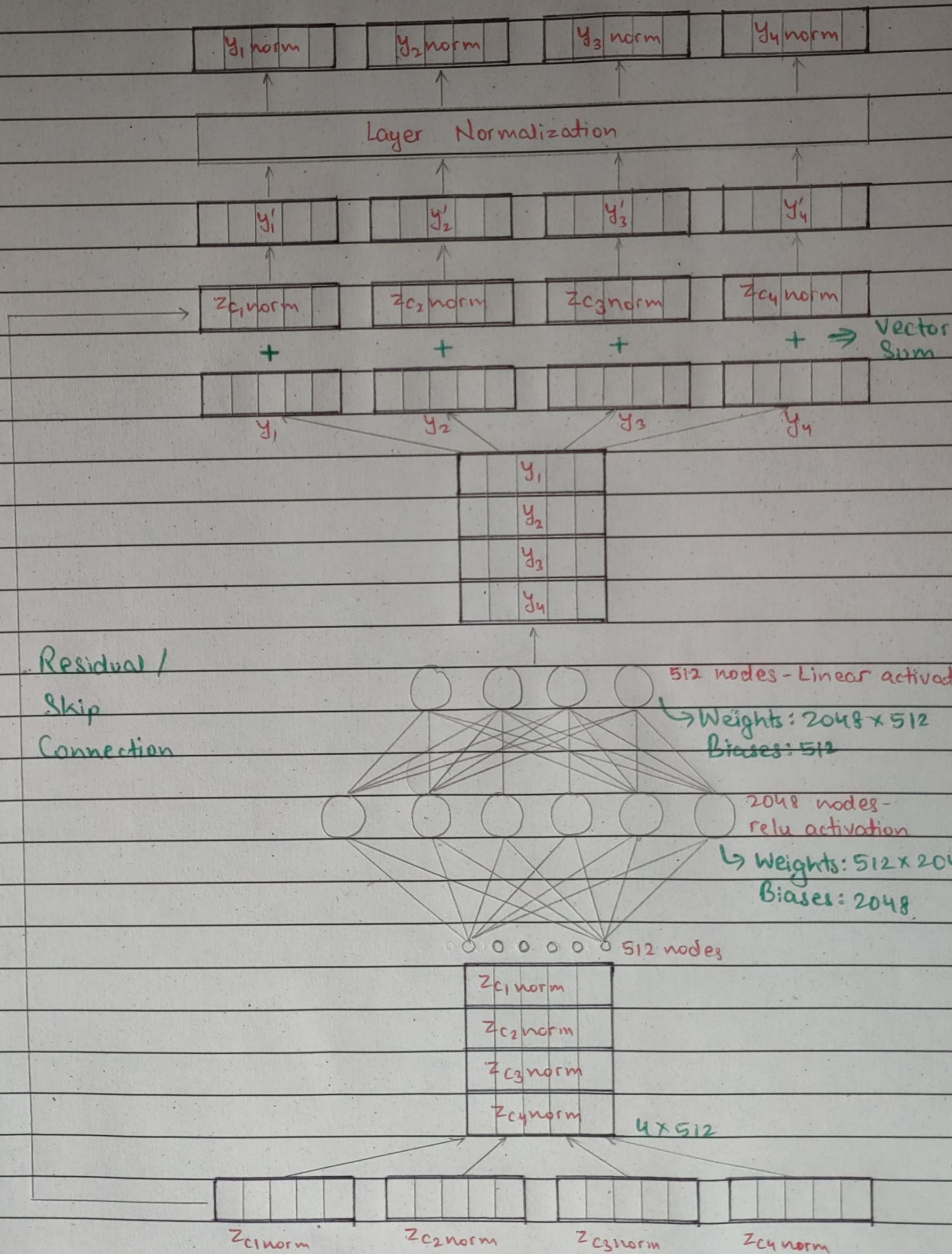
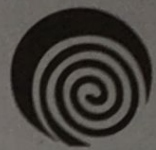




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— a vision beyond —





- * z_{c1norm} , z_{c2norm} , z_{c3norm} & z_{c4norm} from previous stage is taken as input. They are stacked as a 4×512 matrix that is fed into a Feed Forward NN.
- * The NN has 2 layers (excluding input layer). The hidden layer has 2048 neurons all with ReLU activation to capture non-linearity & the output layer has 512 neurons all with Linear activation.
- * Hidden layer has 512×2048 weights & 2048 biases & output layer has 2048×512 weights & 512 biases.
- * The input is of 4×512 so the calculation will be -
$$\text{ReLU}[(4 \times 512) \times (512 \times 2048) + 2048] \quad \text{At hidden layer}$$
$$\Rightarrow \text{ReLU}[(4 \times 2048) + 2048] \rightarrow \text{Resultant matrix}$$
- * For the output layer the input is of 4×2048 dimension, so -
$$\text{Linear}[(4 \times 2048) \times (2048 \times 512) + 512]$$
$$\Rightarrow \text{Linear}[(4 \times 512) + 512] \rightarrow \text{Resultant matrix}$$
- * Through these operations we get the output of original dimensions of 4×512 matrix. This can be broken down into 4 512 dimensional vectors $\rightarrow y_1, y_2, y_3$ & y_4 .
- * Through the residual/skip connection we add original z_{c1norm} , z_{c2norm} , z_{c3norm} & z_{c4norm} to y_1, y_2, y_3 & y_4 . The resultant vectors are y'_1, y'_2, y'_3 & y'_4 .
- * y'_1, y'_2, y'_3 & y'_4 are then passed to Layer Normalization to normalize the embedding values.
- * The output we get are y_{1norm} , y_{2norm} , y_{3norm} & y_{4norm} .
- * y_{1norm} , y_{2norm} , y_{3norm} & y_{4norm} are given as input to the next decoder block & the whole process repeats for 6 decoder blocks.
- * The output of the final decoder block is taken as y_{f1norm} , y_{f2norm} , y_{f3norm} & y_{f4norm} .